AI Usage Declaration and Learning Report

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Subject: HTML & CSS Website Development

AI Usage Declaration

For this assignment, I developed a five-page HTML website. I used ChatGPT as a learning and reference tool to understand HTML5 semantic tags and website structure.
The HTML pages were studied, understood, and modified by me after learning the concepts. I practiced the code and learned the purpose of each HTML tag used in the project.
The external CSS (style.css) was generated with the help of ChatGPT because I did not have sufficient knowledge about choosing color combinations, gradients, layouts, hover effects, Flexbox, CSS Grid, and responsive styling. I studied the CSS code and understood the function of each section before using it in my project.
I have not used AI to avoid learning. Instead, I used it as a reference to improve my understanding of HTML and CSS.

HTML Tags Learned
1. <meta>
Purpose
The <meta> tag provides information about the webpage that is not directly displayed.
Types Used
<meta charset="UTF-8">
Supports different languages and special characters.

<meta name="viewport" content="width=device-width, initial-scale=1.0">
Makes the webpage responsive on mobile, tablet, and desktop devices

2. <nav>

Purpose
Used to create the navigation menu of the website.
Example
<nav>
Home
About
</nav>

3. <main>
Purpose
Contains the main content of the webpage.
Only one <main> element should normally be used on each page.

4. <section>
Purpose
Groups related content together.
Example:

* About College

* Departments

* Gallery

5. <article>

Purpose
Contains independent content that can be understood on its own.
Example:

* Latest News

* Admission Information

6.

Purpose
Emphasizes text.

The browser usually displays the text in italic.

Example

Admissions Open

7. <aside>

Purpose

Contains additional information related to the main content.

Example:

* Admission Notice

* Quick Links

8. <figure>

Purpose
Groups an image together with its caption.
Example

<figure>
<img src="college.jpg"> </figure>

9. <figcaption>
Purpose
Provides a caption or description for the image inside a <figure> element.
Example

<figcaption>Main College Building</figcaption>

10. <footer>
Purpose
Displays information at the bottom of the webpage.
Usually contains:
* Copyright

* Contact details

* Email

* Phone number

11. <small>
Purpose
Displays text in a smaller font size.
Used for:

* Copyright

* Notes

* Disclaimer

12. <thead>

Purpose
Defines the header section of a table.
Contains column headings.
Example
<thead>
<tr>
<th>Department</th>
<th>Duration</th>
</tr>
</thead>

13. <tbody>

Purpose
Contains the main data of the table.
Example

```
<tbody>
<tr>
<td>B.Sc Computer Science</td>
<td>3 Years</td>
</tr>
</tbody>
```

14. <tfoot>

Purpose
Defines the footer section of a table.
Used for totals or summary information.

Example

```
<tfoot>
<tr>
<td colspan="4">Total Departments: 6</td>
</tr>
</tfoot>
```

CSS Learning

The style.css file was generated with the help of ChatGPT because I was not confident in designing attractive webpages and selecting suitable color combinations.

After generating the CSS, I studied and understood each part.
I learned the following CSS concepts:

Yes. Since you are mentioning that only the CSS was AI-assisted, it is a good idea to explain every CSS section in your report. Here is a clear explanation based on the style.css file.

CSS Concepts Learned

1. Universal Selector (*)

Purpose
The universal selector applies styles to all HTML elements.

```
*{
 margin:0;
 padding:0;
```

box-sizing:border-box;
}

Explanation

* margin:0; → Removes the default outer spacing.
* padding:0; → Removes the default inner spacing.
* box-sizing:border-box; → Includes padding and border in the element's total width and height.

2. Element Selector

Purpose
Styles a specific HTML element.
Example:

Body{

font-family:Arial, Helvetica, sans-serif;
background:#f4f4f4;
color:#333;
line-height:1.6;
}

Explanation

* font-family → Changes the font style.
* background → Sets the page background color.
* color → Sets the text color.
* line-height → Increases the spacing between lines.

3. Header Styling
header{

background:linear-gradient(to right,#003366,#0066cc);
color:white;
text-align:center;
padding:25px;

}

Explanation

* background → Creates a gradient background.
* color:white → Makes text white.
* text-align:center → Centers the text.
* padding → Adds space inside the header.

4. Navigation Bar

```
nav{
 background:#222;
display:flex;
justify-content:center;
position:sticky;
 top:0;
}
```

Explanation

* display:flex → Arranges menu items in one row.
* justify-content:center → Centers the menu.
* position:sticky → Keeps the navigation bar visible while scrolling.
* top:0 → Sticks it to the top.

5. Navigation Links

```
nav a{
  color:white;
 text-decoration:none;
 padding:15px 20px;
}
```

Explanation

* color → Text color.
* text-decoration:none → Removes the underline
* padding → Increases the clickable area.

6. Hover Effect

```
nav a:hover{
 background:#00b894;
}
```

Explanation

The :hover pseudo-class changes the menu background color when the mouse pointer moves over it.

7. Main Content

```
main{
padding:20px;
}
```

Explanation

Adds space inside the main content area.

8. Section

```
section{
margin:20px 0;
}
```

Explanation

Adds space above and below each section.

9. Article

```
article{
background:white;
padding:20px;
border-radius:10px;
 box-shadow:0 0 10px gray;
}
```

Explanation

* White background
* Rounded corners
* Shadow effect
* Inner spacing

10. Aside

```
aside{
background:#dfe6e9;
padding:15px;
border-left:5px solid #0984e3;

}
```

Explanation

Creates a sidebar with:

* Light background
* Left colored border
* Inner spacing

11. Figure

```
figure{
 text-align:center;

}
```

Explanation

Centers the image and its caption.

12. Image Styling

```
figure img{
  width:100%;
 max-width:500px;
  border-radius:10px;
}
```

Explanation

* width:100% → Makes the image responsive.
* max-width:500px → Limits the maximum size.
* border-radius → Gives rounded corners.

13. Image Hover

```
figure img:hover{
 transform:scale(1.05);
}
```

Explanation

Enlarges the image slightly when the mouse hovers over it.

14. Gallery Grid

```
.gallery{
  display:grid;
  grid-template-columns:repeat(auto-fit,minmax(250px,1fr));
  gap:20px;
}
```

Explanation

Uses CSS Grid to arrange images automatically into columns with equal spacing.

15. Table

```
table{
  width:100%;
 border-collapse:collapse;  }
```

Explanation

* width:100% → Table uses the full width.
* border-collapse → Merges adjacent borders into a single border.

16. Table Header

```
th{
 background:#0984e3;
 color:white;
}
```

Explanation

Styles the header row with a blue background and white text.

17. Form

```
form{
  background:white;
  padding:20px;
  border-radius:10px;
}
```

Explanation

Makes the form look neat with spacing and rounded corners.

18. Input Fields

```
input,

select,

textarea{
   width:100%;
   padding:10px;
}
```

Explanation

Makes all form controls:

* Equal width
* Comfortable to type in
* Properly spaced

19. Submit Button

```
input[type="submit"]{
    background:#0984e3;
    color:white;

}
```

Explanation

Styles the submit button with a blue background and white text.

20. Footer

```
footer{
  background:#222;
  color:white;
  text-align:center;

}
```

Explanation

Creates a dark footer with centered white text.

21. Class Selector

```
.highlight{
  color:red;

}
```

Explanation

Applies the style to any element with class="highlight".

22. ID Selector

```
#welcome{
   background:#ffeaa7;
}
```

Explanation

Applies the style to the single element with id="welcome".

23. Group Selector

```
h1,h2,h3{
 margin-bottom:15px;   }
```

Explanation

Applies the same style to multiple heading elements.

24. Descendant Selector

```
nav a{
   font-weight:bold;
}
```

Explanation

Styles only the <a> elements that are inside the <nav> element.

25. Position Properties

Relative

```
.relative-box{
position:relative;
 left:20px;
}
```

Moves the element relative to its normal position.

Absolute

```
.absolute-box{
  position:absolute;
  top:10px;
  right:10px;
}
```

Positions the element relative to its nearest positioned parent.

Fixed

```
.fixed-box{
   position:fixed;
   bottom:20px;
   right:20px;
}
```

Keeps the element fixed on the screen even while scrolling.

Sticky

```
position:sticky;
top:0;
```

Keeps the navigation bar at the top after reaching it during scrolling.

26. Flexbox

```
.flex-container{
   display:flex;
   justify-content:space-around;
}
```

Explanation

Arranges child elements in a flexible horizontal layout with equal spacing.

27. Media Query

```
@media(max-width:768px){
 nav{
 flex-direction:column;
}
}
```

Explanation

Makes the website responsive. On screens smaller than 768px, the navigation links are displayed vertically.

Conclusion

By studying the generated CSS, I learned how to style a webpage using selectors, colors, typography, spacing, tables, forms, Flexbox, Grid, positioning, hover effects, transitions, transforms, and responsive design. Although the CSS was initially generated with ChatGPT, I reviewed each rule, understood its purpose, and learned how it affects the appearance and layout of the website.